

# Porcine retinal explants as an ex vivo test system to assess the translational potential of vector-delivered gene therapy strategies in the neuro-retina

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**Purpose:** Inherited retinal diseases (IRD), such as retinitis pigmentosa (RP), significantly contribute to progressive vision loss worldwide, impairing patients' quality of life. Restoring gene function by gene supplementation therapy (GT) or therapeutic gene editing (GE) offers promising treatment strategies. However, clinical translation has proven difficult so far, presumably due to the particular architecture and functional properties of the retina. Here I speculate that retinal explants (RE) derived from genetically modified (GM) pig models provide a precious resource for systematic testing of GT and GE approaches and likewise reduce the number of animal experiments.

**Methods:** Initially, GT and GE strategies were pre-tested in primary kidney cells (PKC) from GM pigs by plasmid transfection. For effective translation, RE from two GM RP pig models were then used to identify suitable vectors for GT and GE and eventually for validating therapies ex vivo. RE were tested for optimal transduction with different vectors with reporter genes such as Green Fluorescent Protein (GFP). Optimized Real-Time PCR assays were then used to evaluate the transcript levels of reporter, GT and GE products. Transduction efficacy was assessed by flow cytometry analysis, quantifying the number of GFP positive cells in the RE. Sanger sequencing enabled the tracking of genomic modifications by GT and GE in RE.

**Results:** So far, experiments in PKC with a humanized USH1C gene, carrying a deleterious R31X mutation, aiming at the reconstitution of the intact reading frame obtained base editing and prime editing efficacies of 20 - 40%. A disruptive GE strategy, making use of non-homologous end-joining, on a RHO-P347L mutation showed higher efficiency when two sgRNAs were combined (30 - 50%). However, specificity of allele-specific disruptive GE in PKC remained questionable. In RE I tested different adeno-associated viruses and virus like particles with reporter genes and found transduction generally to increase in a dose dependent manner. Interestingly, I observed that the genotype of the RE, the vector serotype, and the integrity of the tissue all influenced transduction.

**Conclusions:** GM pigs are of great value for translational research on IRDs, especially when the gap between the cellular level and in vivo experiments can be bridged through systematic testing in RE. Next, we will introduce USH1C and RHO-P347L GE into RE from GM pigs to address outstanding questions.

**Keywords:** Organotypic retina culture, Retinitis pigmentosa, Gene therapy, Adeno-associated viruses, Translational medicine